

Program

```
#include <stdio.h>
#include <time.h>
int max(int a, int b)
{
    if (a > b)
        return a;
    return b;
}
```

```
int main()
{
    clock_t start, end;
    int P[100], W[100], M, n, Table[100][100];
    printf("Enter the number of compartments items: ");
    scanf("%d", &n);
    printf("Enter maximum capacity of the knapsack: ");
```

```
    for (int i = 1; i <= n; i++)
```

```
    {
        printf("Enter the weight & profit of item %d", i);
        scanf("%d %d", &W[i], &P[i]);
    }
```

```
    start start = clock();
```

```
    for (int i = 0; i <= n; i++)
```

```
    {
        for (int j = 0; j <= M; j++)
```

```
        {
            if (i == 0 || j == 0)
```

```
                Table[i][j] = 0;
```

```
else if (j < w[i])
{
```

```
    Table[i][j] = Table[i-1][j];
```

```
    }
else
{
```

```
    Table[i][j] = max( Table[i-1][j],
                       P[i] + Table[i-1][j-w[i]] );
```

```
    }
```

```
} end = clock();
```

```
printf("Maximum profit made: %.d",
       Table[n][m]);
```

```
printf
```

```
double time = (double)(end - start) /
               CLOCKS_PER_SEC;
```

```
printf("Execution time: %.f",
       time);
```

```
}
```

Output:

Enter number of items: 3

Enter maximum capacity of knapsack: 4

Enter weight & profit of item 1: 2 6

Enter weight & profit of item 2: 2 6

Enter weight & profit of item 3: 1 8

Maximum profit made: 34

Execution time: 0.000000

Vostok